

Lab Report: Week 8 - Interference and Diffraction of Light

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Pre-Lab Question

From equation 3,

$$y_{dark} = L \left(\frac{m + \frac{1}{2}}{d} \right), \quad m = 0, \pm 1, \pm 2, \dots$$

The first dark fringe on either side of the central maximum corresponds to $m = 0$ (positive side) and $m = -1$ (negative side), which are equidistant from the centre

$$y_{\text{dark}}(m = 0) = L \frac{\frac{1}{2}\lambda}{d}, \quad y_{\text{dark}}(m = -1) = -L \frac{\frac{1}{2}\lambda}{d}$$

So the separation between the two symmetric first dark fringes is:

$$\Delta y = 2 y_{\text{dark}}(m = 0) = \frac{L\lambda}{d}$$

Substituting values: $L = 1 \text{ m}$, $\lambda = 650 \text{ nm} = 650 \times 10^{-9} \text{ m}$, $d = 0.3 \text{ mm} = 3 \times 10^{-4} \text{ m}$,

$$\Delta y = \frac{(1)(650 \times 10^{-9})}{3 \times 10^{-4}} = 2.1667 \times 10^{-3} \text{ m}$$

$$\boxed{\Delta y \approx 2.17 \text{ mm}}$$